

The preCICE system regression tests

Plain language summary

The software library preCICE allows programmers and end users to combine simulation models with minimal effort. Integrations with simulation codes are developed by multiple people in various places, and useful simulations can only be executed by combining multiple of theses codes. To test that development changes do not affect the numerical results, we create reproducible environments from a common description and for the versions of interest, connect them, and check the results at the interface between the individual simulations. With this mechanism in place, the community can now shape their contributions in a way that can be continuously tested and easily maintained.

Coupling via preCICE

Add calls to the preCICE library into each code, or use provided adapters. Start each code in a separate process (e.g., two terminals).

OpenFOAM

FEniCS

```
import preCICE
v = preCICE.set_mesh_vertices(positions)
preCICE.initialize()
while preCICE.is_coupling_ongoing():
    solve_as_usual()
    preCICE.write_data("temperature", v)
    preCICE.advance()
    preCICE.read_data("heat-flux", v)
```

A citable ecosystem of components

citable preCICE Distribution (on DaRUS ^[2])			
Website + Documentation		Nix pkgs	VM
Tutorials			
OpenFOAM	CalculiX	FEniCS	deal.II
SU2	DUNE	DuMuX	MBDyn
Python	Julia	Matlab	Rust
preCICE (C++ core, C & Fortran bindings)			

Community contributions

preCICE store

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Developing guidelines

☐ Metadata, citing

☐ Tests available

☐ Reference results

☐ Executing via scripts

preCICE conforming

Gold level

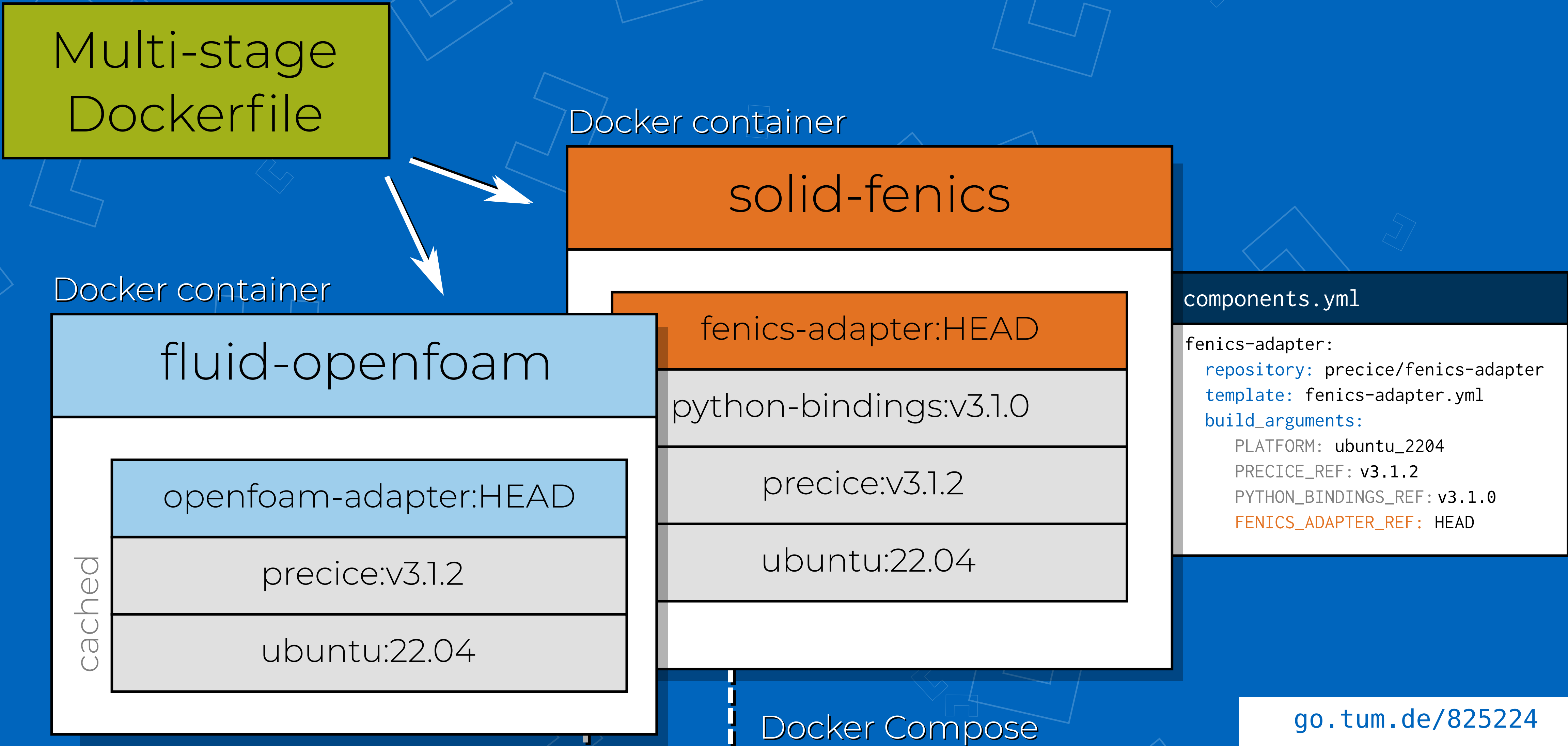
preCICE conforming

Silver level

preCICE conforming

Bronze level

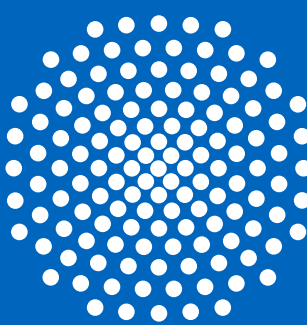
Ensuring your simulations still run correctly



[1] G. Chourdakis, K. Davis, B. Rodenberg, M. Schulte, F. Simonis, B. Uekermann, et al., preCICE v2: A sustainable and user-friendly coupling library [version 2; peer review: 2 approved] Open Res Europe 2022, 2:51, <https://doi.org/10.12688/openreseurope.14445.2>

[2] J. Chen, G. Chourdakis, I. Desai, C. Homs-Pons, B. Rodenberg, D. Schneider, F. Simonis, B. Uekermann, K. Davis, A. Jaust, M. Kelm, N. Kotarsky, H. Kschidock, D. Mishra, M. Mühlhäußer, T. P. Schrader, M. Schulte, V. Seitz, J. Signorelli, G. van Zwieten, N. Vinnitschenko, T. Vladimirova, L. Willeke, E. Zonta (2024) preCICE Distribution Version v2404.0. <https://doi.org/10.18419/darus-4167>, DaRUS, V1.

[3] G. Chourdakis, V. Seitz, B. Uekermann (2024) System regression tests for the preCICE coupling ecosystem Proceedings of the deRSE24, Electronic Communications of the EASST (in review)



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Reference results and comparisons

Stored next to the cases

Git Large File Storage

git-lfs.com

VTK compared with

FieldCompare

gitlab.com/dglaeser/fieldcompare

flow-over-heated-plate/
fluid-openfoam/
solid-fenics/
solid-nutils/
reference-results/
fluid-openfoam_solid-fenics.tar.gz
fluid-openfoam_solid-nutils.tar.gz

One directory for each example
One directory for each code
Multiple code options
Reference results versioned
One file for each combination

fluid-openfoam	preCICE	solid-fenics
OpenFOAM output .foam, logs	mesh exports (.vtu) iterations.log	FEniCS output .vtk, logs
specific to OpenFOAM, controlled by others	standard formats (VTU, CSV)	specific to FEniCS, controlled by others

Metadata, tests, and test suites

metadata.yaml (precice/tutorials/flow-over-heated-plate/)

name: Flow over heated plate
path: flow-over-heated-plate
participants:
- Fluid
- Solid

cases:
fluid-openfoam:
participant: Fluid
directory: ./fluid-openfoam
run: ./run.sh
component: openfoam-adapter

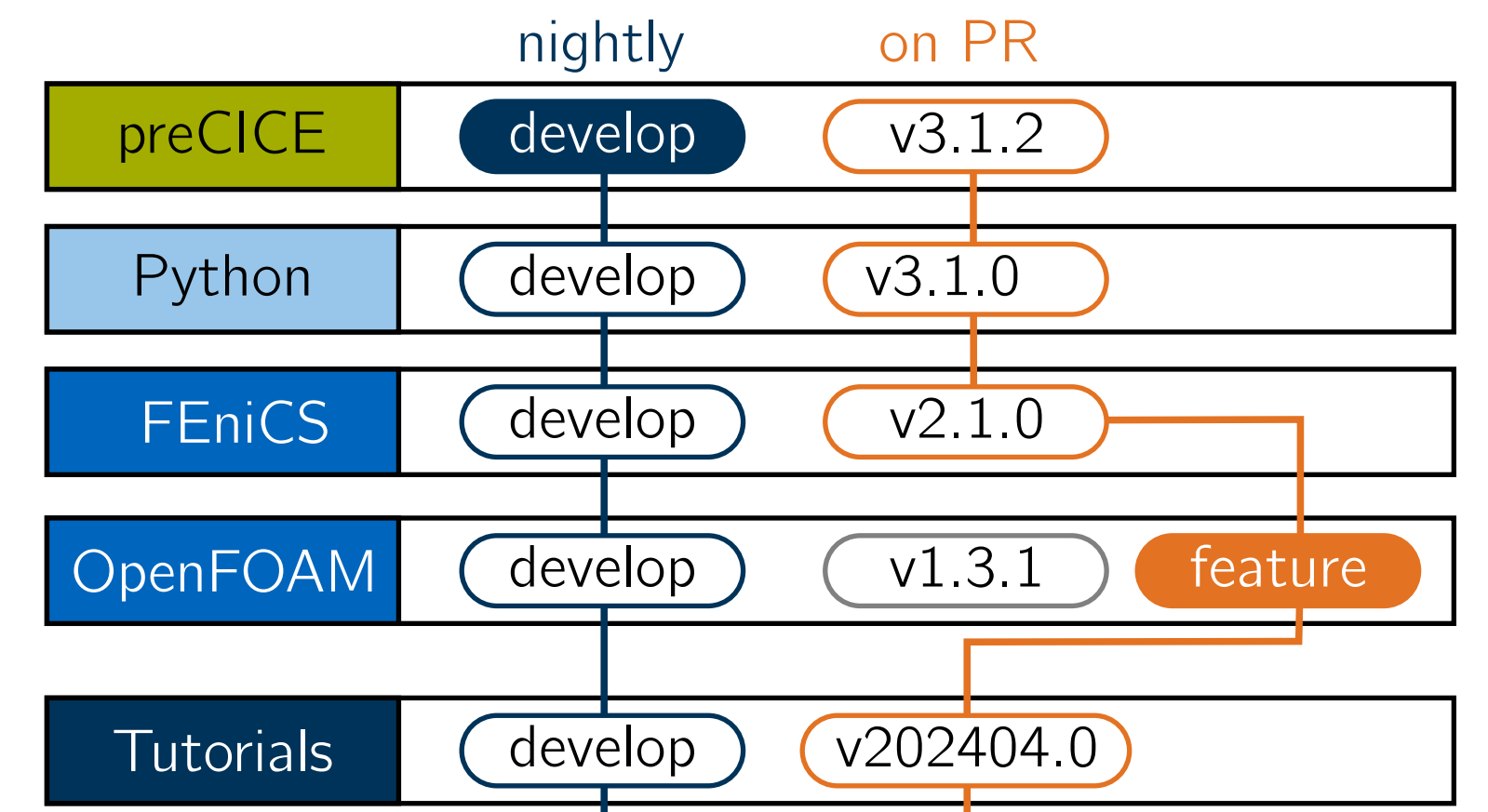
tests.yaml (precice/tutorials)

test_suites:
openfoam_adapter_pr:
tutorials:
- path: flow-over-heated-plate
case_combination:
- fluid-openfoam
- solid-fenics
reference_result:
fluid-of-solid-fenics.tar.gz

solid-fenics:
participant: Solid
directory: ./solid-fenics
run: ./run.sh
component: fenics-adapter

Multiple repos, multiple perspectives

Partitioned approach makes CI complex. Tests can run locally or on GitHub Actions, triggered manually or by pull requests.



Can we publish a new release? Can we merge the PR?

openfoam-adapter-workflow.yaml (precice/openfoam-adapter)

jobs:
gather-refs: (map from branch to latest commit hash)
run-system-tests:
if: \${github.event.label.name == 'trigger-system-tests'}
uses: precice/tutorials/.github/workflows/workflow.yaml@develop
with:
suites: openfoam_adapter_pr
build_args: OPENFOAM_ADAPTER_REF:\${github.event.pr.head.sha}
systests_branch: develop

workflow.yaml (precice/tutorials)

jobs:
run_testsuite:
runs-on: [self-hosted, linux, x64, precice-tests-vm]
steps:
- name: Run tests
run: |
cd tools/tests
python systemtests.py
--build_args=\${{inputs.build_args}}
--suites=\${{inputs.suites}}